D 53686 **(Pages : 3)** **Name.....**

Reg. No.....

FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2023

Physics/Applied Physics (Common for Double Main)

PHY 1B 01/APH 1B 01/PHY 1B 21—MECHANICS—I

(2020—2023 Admissions)

Time : Two Hours Maximum : 60 Marks

The symbols used in the question paper have their usual meanings

Section A (Short Answer Type)

Answer **all** *questions in two* **or** *three sentences, each correct answer carries a maximum of* 2 *marks.*

1. Explain Fictitious force.
2. State Newton's third law of motion with one example.
3. With proper examples define contact forces.
4. What are the fundamental forces in nature ? Compare their strengths.
5. Explain the gravitational force of a sphere.
6. State and explain the work energy theorem.
7. Define moment of inertia. How is it related to angular momentum ?
8. Draw the energy diagram for a harmonic oscillator and explain.
9. Illustrate the working of Drum major's baton.
10. What are conservative forces ? Give examples.
11. Define power with its various units.
12. State and explain parallel axis theorem.

(Ceiling - 20)

Turn over

Section B (Paragraph / Problem Type)

Answer **all** *questions in a paragraph of about half a page to one page, each correct answer carries a maximum of* 5 *marks.*

13. A 5 kg. mass moves under the influence of a force $F = \left(4t^2\hat{i} - 3t\hat{j}\right)$ N, where t is the time in seconds. (IN = 1 Newton). It starts at rest from the origin at $t = 0$. Find :

    (a) Its velocity.

    (b) Its position.

    (c) $r \times v$, for any later time.

14. A Mass m is whirled on the end of a string length **R**. The motion is in a vertical plane in the gravitational field of the earth. The forces on m are the weight **W** down, and the string force T towards the centre. The instantaneous speed is **v**, and the string makes angle ϑ with the horizontal. Find the **T** and the tangential acceleration at this instant.

15. Solve the equation of simple harmonic motion.

16. A uniform rope of mass ***m*** and length ***l*** is attached to a block of mass ***M***. The rope is pulled with force ***F***. Find the tension at distance ***x*** from the end of the rope. Neglect gravity

17. Write a note on small oscillation in a bound system

18. Show that if the total force on a system of particles is zero, the torque on the system is the same around all origins..

19. Three freight cars each of mass M are pulled with force F by a locomotive. Friction is negligible. Find the force on each car ?

(Ceiling - 30)

Section C (Essay Type)

Essays-Answer in about **two pages**, *any one question.*

The question carries 10 *marks.*

20. Define potential energy.

    (a) Obtain Potential energies of a uniform force field.

    (b) Obtain Potential energy of a central force ; and

    (c) Obtain the Potential energy of the Three-dimensional Spring Force.

21. State Newton's laws of motion. Apply Newtons laws to find the accelerations of two astronauts of masses M_A and M_B pulling on either ends of a rope of negligible mass.

(1 × 10 = 10 marks)